Earthquakes and Tsunamis

Knowledge document 036 of 100

Introduction to Earthquakes and Tsunamis

An earthquake is the sudden shaking of the ground caused by a rapid release of energy in the Earth's crust. A tsunami is a series of very long ocean waves, usually generated by a large underwater earthquake, volcanic eruption or landslide, that can flood coastlines. The word tsunami comes from Japanese and means "harbour wave". Earthquakes and tsunamis are among the deadliest natural disasters, and understanding them helps communities prepare and save lives. Scientists who study earthquakes are called seismologists, and the science is called seismology.

What Causes an Earthquake?

Most earthquakes are caused by the movement of tectonic plates. The Earth's rigid outer shell, the lithosphere, is broken into about a dozen large plates and several smaller ones, which move slowly, a few centimetres a year, over the hot, soft mantle below. Where plates collide, slide past each other or pull apart, enormous stress builds up along cracks in rocks called faults. When the stress exceeds the strength of the rock, the rock suddenly slips and energy is released as seismic waves, the shaking that we feel. This idea is called the elastic rebound theory. The point underground where the rupture starts is called the focus or hypocentre, and the point on the surface directly above it is the epicentre. Earthquakes can also be caused by volcanic activity, mining, large reservoirs and nuclear explosions, though these are less common. Most earthquakes occur along plate boundaries, such as the Pacific Ring of Fire.

Seismic Waves

P-waves (primary waves) are compressional, travel fastest and arrive first. S-waves (secondary waves) move the ground sideways, are slower and cannot travel through liquids, which helped scientists discover that the Earth's outer core is liquid. Surface waves travel along the ground and cause most of the damage. The instrument that records ground motion is called a seismograph, and its record is a seismogram.

What Is the Richter Scale?

The Richter scale is a scale for measuring the magnitude, or size, of an earthquake. It was developed in 1935 by the American seismologist Charles F. Richter, working with Beno Gutenberg at the California Institute of Technology. The scale is logarithmic, which means that each whole-number increase represents ten times larger ground motion amplitude, and about 32 times more energy released. For example, a magnitude 7 earthquake produces ten times larger ground shaking than a magnitude 6 earthquake and releases about 32 times more energy. Earthquakes below magnitude 3 are usually not felt, those of magnitude 4 to 5 can be felt and cause light damage, those from 6 to 7 can cause serious damage, and those of magnitude 8 or higher are great earthquakes that can destroy whole regions. Today, scientists mainly use the moment magnitude scale (Mw), which works better for very large earthquakes, though the media often still call it the Richter scale. A different scale, the Modified Mercalli Intensity scale, with levels I to XII, describes the effects and damage at a place instead of the earthquake's size.

Earthquakes in India

India is prone to earthquakes because the Indian plate moves northward and collides with the Eurasian plate, forming the Himalaya. The Bureau of Indian Standards divides India into four seismic zones, II, III, IV and V, with Zone V being the most severe; it includes the north-eastern states, parts of Kashmir, Uttarakhand, and the Rann of Kutch. Most of Tamil Nadu, including Chennai, lies in the lower-risk Zones II and III. Notable Indian earthquakes include the Latur earthquake of September 1993 in Maharashtra (magnitude 6.2, about 10,000 deaths), the Bhuj earthquake in Gujarat on 26 January 2001 (magnitude 7.7, about 20,000 deaths), the Kashmir earthquake of 2005 and the Nepal earthquake of April 2015, which shook north India.

What Is a Tsunami and How Does It Form?

Most tsunamis form when a large undersea earthquake, typically at a subduction zone where one plate dives beneath another, suddenly lifts or drops the seafloor, displacing a huge volume of water above it. This starts waves that spread outward in all directions. In deep ocean, a tsunami wave may be only a metre high but moves at the speed of a jet aeroplane, about 700 to 800 kilometres per hour, and has a wavelength of hundreds of kilometres, so ships may not notice it. As the wave reaches shallow water near the coast, it slows down, and its height grows rapidly, sometimes to 10 metres or more, striking the coast as a surging wall of water with great force. A sign of an approaching tsunami can be the sea suddenly retreating far from the shore, in which case people should move immediately to high ground.

What Happened in the 2004 Indian Ocean Tsunami?

On 26 December 2004, a massive undersea earthquake of magnitude about 9.1 to 9.3 struck off the west coast of northern Sumatra, Indonesia, along the Sunda megathrust, one of the largest earthquakes ever recorded. The fault ruptured for about 1,200 to 1,300 kilometres. The resulting tsunami travelled across the Indian Ocean within hours, with waves as high as 30 metres near Sumatra, killing about 230,000 people in 14 countries, including Indonesia, Sri Lanka, India and Thailand, and even reaching the coast of Africa. In India, around 10,000 to 12,000 people died, mostly in Tamil Nadu, Andhra Pradesh, Kerala, Puducherry and the Andaman and Nicobar Islands. Tamil Nadu suffered the worst; Nagapattinam district alone lost about 6,000 people, and Chennai's Marina beach, Cuddalore and Kanyakumari were also hit, with fishing villages destroyed. Millions lost their homes and livelihoods.

Warning Systems and Safety

After 2004, India established the Indian Tsunami Early Warning Centre at the Indian National Centre for Ocean Information Services (INCOIS) in Hyderabad, which began operating in October 2007. It uses seismic stations, deep-ocean pressure sensors called tsunami buoys and tide gauges to issue warnings within minutes. The Disaster Management Act of 2005 created the National Disaster Management Authority (NDMA). During an earthquake, the advice is "Drop, Cover and Hold On": drop to the ground, take cover under a sturdy table, and hold on until shaking stops, staying away from windows and not using lifts. After a strong coastal earthquake, move to high ground and do not wait for an official warning. Buildings can be made earthquake-resistant through flexible designs and codes such as IS 1893.

Challenges and Future

Earthquakes cannot be predicted precisely in time, though early warning systems can give seconds of notice. Future work includes better building codes, AI-assisted seismology, and community drills. Mangroves and coastal forests can reduce tsunami damage.

Quick Facts

- Earthquakes are caused by the sudden release of stress along faults, mostly at tectonic plate boundaries.
- The Richter scale, developed by Charles Richter in 1935, is logarithmic; each whole number means ten times more shaking.
- The 2004 Indian Ocean tsunami occurred on 26 December 2004 after a magnitude 9.1 to 9.3 earthquake off Sumatra and killed about 230,000 people.
- Nagapattinam in Tamil Nadu was the worst hit Indian district in the 2004 tsunami.

- India's Tsunami Early Warning Centre is at INCOIS, Hyderabad.